

Coaching Agile Software Projects

Tutorial proposal – ICSE 2010

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ABSTRACT

Our tutorial for ICSE 2010 focuses on coaching agile software teams.

Based on eight years of experience guiding *agile* software projects in the academia and industry, we focus in the tutorial on a coaching framework for agile software projects. The tutorial participants become familiar with coaching practices and gain experience with some of the practices.

The tutorial has two main parts. In the first part, we present the coaching framework, including the goals, structure, and guiding principles. In the second part, we focus on the following central themes in agile development processes which, we suggest, are appropriate to be included in such a coaching framework: Teamwork and collaboration, time and measures, learning and reflection, and change and leadership.

Since 2003, we have facilitated this tutorial and similar ones in different settings (industry, academia, conferences).

The coaching framework, as well as the themes, case studies, and analysis approach presented in the tutorial, are summarized in our book *Agile Software Engineering* published by Springer in 2008 (Hazzan and Dubinsky, 2008).

Categories and Subject Descriptors

D.2.9 [Software]: Software Engineering – *management*.

General Terms

Management, Measurement, Human Factors, Standardization, Languages, Legal Aspects.

Keywords

Software engineering, coaching, agile software development, software project management, retrospective, learning processes.

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1. PRESENTES

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Dr. Orit Hazzan is an associate professor at the Department of Education in Technology and Science of the Technion – Israel Institute of Technology. In May 2004, she published her first book *Human Aspects of Software Engineering*, co-authored with the late Jim Tomayko. Her second book – *Agile Software Engineering* – co-authored with Yael Dubinsky, was published by Springer in 2008. She is also a consultant for several software projects in the Israeli software industry. She presents her research at computer science and software engineering education conferences, and conferences on software engineering in general and agile software development in particular.

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Dr. Yael Dubinsky is affiliated with the Software and Services group in IBM Haifa Research Lab. For more than ten years she is also the instructor of project-based courses in the Department of Computer Science, Technion - IIT. Her research interests involve aspects in software engineering and information systems. Yael has a significant experience with guiding agile implementation processes in the industry and academia. She has experience in organizing workshops and tutorials since 2005. Her book on *Agile Software Engineering*, which she co-author with Orit Hazzan, was published by Springer at 2008.

2. DURATION

Half day

3. GOAL AND OBJECTIVES

The overall goal of the tutorial is to introduce and practice a coaching approach and practices for agile project development. Specifically, the objectives of the tutorial are:

- The tutorial participants learn how to initiate and organize the development process of agile projects;
- The tutorial participants realize how to inspire the agile approach while coaching software teams,;

- The tutorial participants apprehend how to improve the professional skills of software engineers.

This tutorial would be of interest and benefit to ICSE 2010 attendees since it is important, suitable and timely to introduce agile software development into software engineering projects. Further, though there are books and tutorials about what agile is, our tutorial (based on our book) provides a framework to learn the agile approach along the development process.

4. SCOPE

Intended audience: This tutorial is relevant for team leaders and software practitioners who are interested in coaching agile software projects. It is also relevant for instructors and coaches in academia who are interested in implementing agile practices as part of their teaching.

Level: From the coaching perspective, the tutorial is basic; however, basic acquaintance with agile software development is recommended.

5. TEACHING METHOD

We use a hybrid approach of lectures and interactive sessions. The interactive parts are based on group work, guided discussions and reflections.

6. SUMMARY AND CONTENTS

The agile approach for software development has been applied more and more extensively since the mid nineties of the 20th century. Though there are only about ten years of accumulated experience using the agile approach, it is currently conceived as one of the mainstream approaches for software development.

In the tutorial we examine and experience the following coaching principles we employ in different coaching frameworks of agile projects:

1. Inspire the nature of the agile software development
2. Let the learners experience the principles of agile software development
3. Explain while doing
4. Elicit reflection on experience
5. Elicit communication
6. Establish diverse teams
7. Assign roles to team members
8. Manage time
9. Be aware of abstraction levels
10. Inspire a process of an on-going and gradual improvement
11. Use metaphors or "other world's concepts"
12. Increase cognitive and social awareness needed for the implementation of agile software development
13. Listen to participants' feelings towards agile software development
14. Emphasize the relevance of agile software development in the context of the software world

These coaching guidelines are examined based on the active participation of the participants in different activities related to

coaching software teams in the agile approach. The activities focus on the following themes:

- *Teamwork and collaboration.* We analyze situations and conditions in software development environments that promote teamwork and collaboration and examine how they support agile software development processes.
- *Time and measures.* We describe how measures can be integrated in agile projects, as well as time management techniques derived from the agile approach.
- *Learning and reflection.* We analyze software development processes as learning processes and illustrate how reflection can be used as one vehicle supporting these learning processes.
- *Change and leadership.* We describe ways to cope with changes in software projects and how leadership is related to change management in agile software development environments.

Table 1: Tutorial Timetable

Duration	Theme
5 minutes	Tutorial introduction
15 minutes	Coaching framework and principles
1.5 hours	Activities will be facilitated with respect to the following themes. Roles Assessment Metaphors
15 minutes	Break
1.5 hours	Activities will be facilitated with respect to the following themes. Time management: Change management Reflection:
10 minutes	Tutorial summary

Each of the discussed agile ideas and themes is reviewed in the tutorial through the following three perspectives:

- The **Human** perspective, which includes cognitive and social aspects, and refers to learning and interpersonal processes between teammates, customers, and management.
- The **Organizational** perspective, which includes managerial and cultural aspects, and refers to software project management and control.
- The **Technological** perspective, which includes practical and technical aspects, and refers to design, testing, and coding, as well as to integration, delivery, and maintenance of software products.

Specifically, we explain and analyze how the explicit attention that agile software development gives these three perspectives and their inter-connections, helps it cope with challenges of software projects.

7. REFERENCES

- [1] Hazzan, O. and Dubinsky, Y. (2008). *Agile Software Engineering*, Springer.